Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 5 – Software Testing Documentation

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V. Software Testing Documentation

1. Scope of Testing

Level	In-Charge	Inputs/Time	Focus	Acceptance Criteria
Unit Testing	Developers	Individual modules during development.	Business logic, validation and edge cases.	All unit tests pass; no critical bugs.
Integration Testing	All Members	After module integration.	APIs, data flow and external services.	Modules integrate correctly; no major errors.
Interface Testing	All Members	After UI implementation.	Role-based navigation, forms and messages.	UI works correctly; no broken navigation.
End-to-End Testing	All Members	Before release, after integration.	Onboarding, ingestion, detection, investigation and subscription workflows.	Core flows meet agreed requirements; no unresolved critical defects.

2. Test Strategy

2.1 Testing Types

Type	Focus
Functional	Verify approved workflows, input validation and expected results.
Security and Authorization	Check role permissions, tenant isolation and sensitive-data protection.
Reliability	Check duplicates, data preservation, retries and AI fallback.
Usability	Check clear navigation, messages, alerts and investigation information.
Regression	Rerun affected tests after changes and defect fixes.

2.2 Test Levels

Level	Approach
Unit	Test isolated logic with normal, invalid and boundary inputs.
Integration	Test module/API interactions with a test database and controlled external services.
Interface	Manually check role-based screens, navigation and validation.
End-to-End	Run complete business journeys on a known test build; record actual results and evidence.

2.3 Supporting Tools

Tool	Purpose
JUnit 5 / Maven	Run backend unit and integration tests.
ArchUnit / Testcontainers	Check module boundaries and database integration.
API client / Browser developer tools	Check API requests, UI behavior and workflow results.
AI test double / Payment sandbox / Notification sink	Control external responses and failure scenarios.
Excel / Git	Maintain test cases, results, evidence links and synthetic test data.

3. Test Plan

3.1 Human Resources

No.	Full Name	Student Code	Role	Responsibilities
1	Lê Hồng Đức	SE182106	Leader	Planning, assign feature groups to test. Write tests and test reports.
2	Huỳnh Khả Tú	SE182134	Member	Write tests and test reports.
3	Nguyễn Minh Huy	SE181524	Member	Write tests and test reports.
4	Nguyễn Thị Diệu Vi	SE181505	Member	Write tests and test reports.

3.2 Test Environment

Component	Planned Setup	Version / Details
Test Build	SME-FraudGuard test deployment	[Build / Commit]
Operating System	Test machine / server	[OS / Version]
Browser	Browser for interface and end-to-end tests	[Browser / Version]
Frontend	Next.js / React / Node.js / npm	[Versions]
Backend	Java / Spring Boot / Maven	[Versions]
Database	Isolated PostgreSQL test database	[Version]
AI Service	Controlled Normal, Anomaly and failure responses	[Model / Service Version]
External Services	Payment sandbox; notification sink	[Provider / Version]
Test Data	Synthetic data; separate workspaces A and B	[Dataset Version]
Environment Record	Deployment address and execution date	[Test URL]; [dd/mm/yyyy]

3.3 Test Milestones

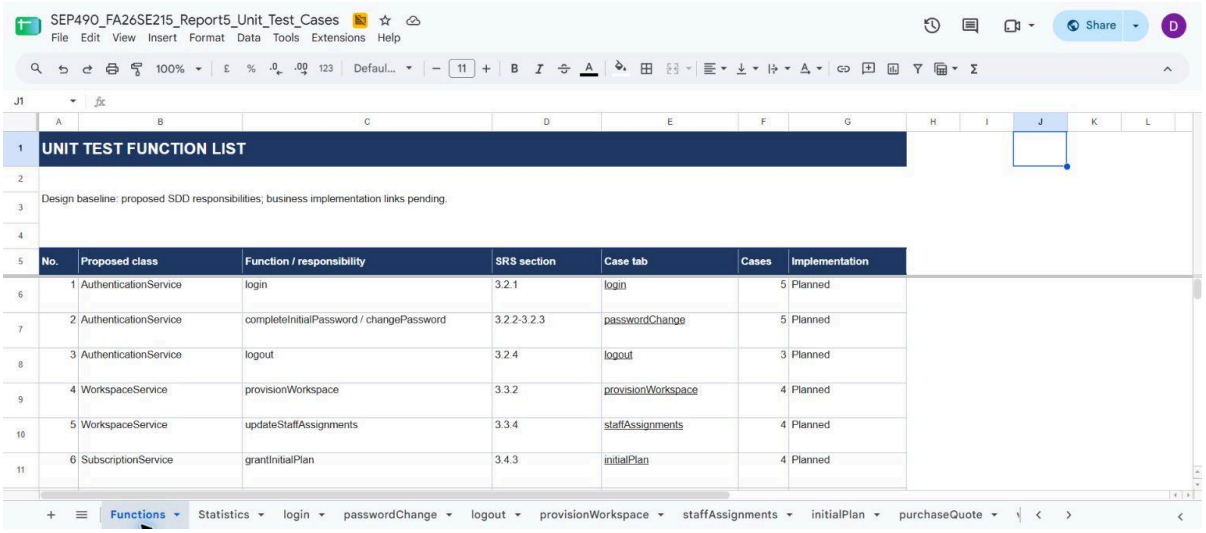
Milestone	Start Date	End Date	Deliverable
Test Preparation	[dd/mm/yyyy]	[dd/mm/yyyy]	Test cases, synthetic data and test environment.
Unit Testing	[dd/mm/yyyy]	[dd/mm/yyyy]	Unit test results for completed modules.
Integration and Interface Testing	[dd/mm/yyyy]	[dd/mm/yyyy]	Integration/UI results and defect records.
End-to-End Testing	[dd/mm/yyyy]	[dd/mm/yyyy]	Core workflow results and evidence.
Regression and Reporting	[dd/mm/yyyy]	[dd/mm/yyyy]	Retest results and consolidated test report.

4. Test Cases

4.1 Unit Test Cases

SEP490_FA26SE215_Report5_Unit_Test_Cases

164 designed case specifications in 33 function matrices. Class/function references follow the proposed SDD and will be aligned with implemented code. All cases are Not Run.



SEP490_FA26SE215_Report5_Unit_Test_Cases

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UNIT TEST FUNCTION LIST

Design baseline: proposed SDD responsibilities; business implementation links pending.

No.	Proposed class	Function / responsibility	SRS section	Case tab	Cases	Implementation
1	AuthenticationService	login	3.2.1	login	5	Planned
2	AuthenticationService	completeInitialPassword / changePassword	3.2.2-3.2.3	passwordChange	5	Planned
3	AuthenticationService	logout	3.2.4	logout	3	Planned
4	WorkspaceService	provisionWorkspace	3.3.2	provisionWorkspace	4	Planned
5	WorkspaceService	updateStaffAssignments	3.3.4	staffAssignments	4	Planned
6	SubscriptionService	grantInitialPlan	3.4.3	initialPlan	4	Planned

Functions Statistics login passwordChange logout provisionWorkspace staffAssignments initialPlan purchaseQuote

Figure 5.4.1. Unit Test Case Catalogue (excerpt).

4.2 System Test Cases

SEP490_FA26SE215_Report5_System_Test_Cases

191 designed cases in 18 feature groups. Detailed procedures, expected results and test-data profiles are maintained in the linked workbook. All cases are Not Run.

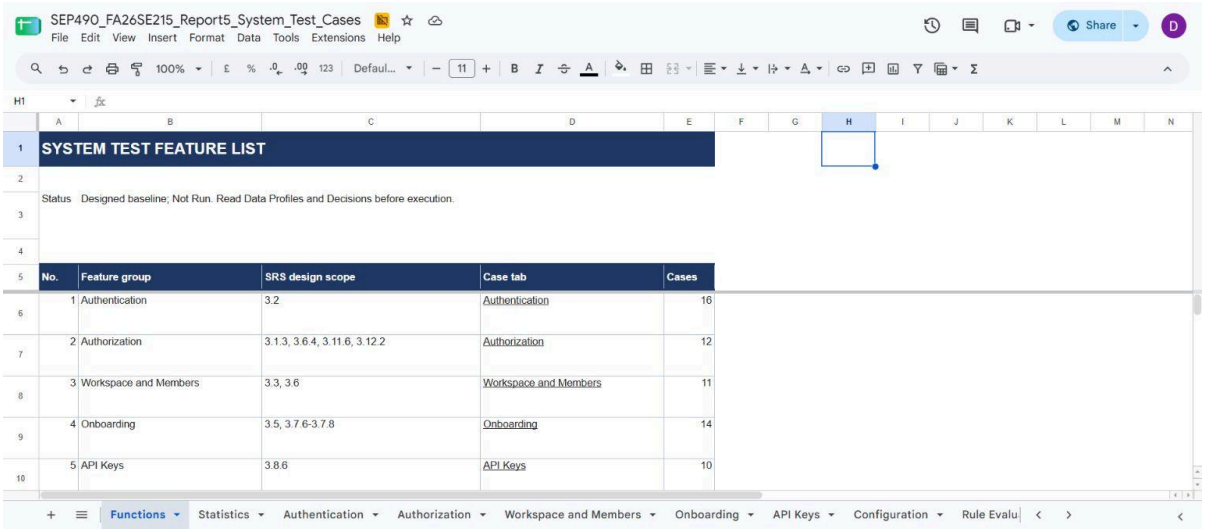


Figure 5.4.2. System Test Case Catalogue (excerpt).

5. Test Reports

5.1 Unit Test Report

5.2 System Test Report